# Case Assignment 3

# BAN 630 – Optimization for Analytics


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# Marico case

**a. Explore what can be potential objectives in this case. Order them by importance.**

The potential objectives in the order of importance could be:

1. **Minimize Freight Costs:**

   In the entire case study Freight costs have been the primary contributor to distribution expenses therefore optimizing primary and secondary freight costs to reduce transportation expenses could be a primary objective for the management of Marico Ltd to consider.

2. **Optimize Warehouse Count:**

   More warehouses increase service levels but raise costs like rent, labor, operations. Therefore, finding the optimal number of warehouses to balance service levels with costs should be of importance.

**b. What should Arun go with, 22 warehouses or 24 warehouses? Why?**

We recommend Arun go with 24 warehouses because it offers better service levels, especially in key regions like the North and West, while also reducing freight costs through more localized service. Although the monthly rent is higher at ₹9,360,000 compared to ₹8,580,000 for 22 warehouses, the latter scenario comes with ₹100,000 more in freight costs.

Additionally, one important reason for us to consider 24 warehouses is because this way there is better support for potential demand growth, making it a more strategic long-term choice rather than cutting down to 22 warehouses where to meet the potential demand growth of future you would need to set up more warehouses again.

So, we would recommend Arun to go with 24 warehouses.

**c. Try and answer the questions bothering Arun:**

**Are 22 warehouses more cost-effective than 24?**
No, while 22 warehouses have lower rental costs, they come with higher freight expenses and a potential drop in service levels, which could lead to lost sales and fulfillment challenges.

**Should future demand growth be factored in?**
Yes, demand growth should be considered, as 24 warehouses offer greater flexibility and scalability to accommodate future increases.

**Can 22 warehouses handle demand elasticity?**
Unlikely, as rising demand from modern trade and e-commerce may strain the 22-warehouse setup, whereas 24 warehouses provide better responsiveness to demand changes.

**2. Simon's Mall**

Simon can maximize its rental income from the mall by allowing the shops as below to be occupied in the mall.

```
Optimal Solution Found:
Jewelry Stores: 2
Shoe Stores: 3
Department Stores: 2
Book Stores: 2
Clothing Stores: 2
Maximum Rental Income: $120000.00
```

The rent that the mall then receives would be as below.

```
🏬 Simon's Mall Rent Calculation:
Total Profit from Stores: $1,170,000.00
Rent Received (7.5% of Profit): $87,750.00
```

However if we increase the maximum limit of the stores by 1 each, then the stores and rent would be as below.

```
🔍 Sensitivity Analysis - Increased Max Limits by 1
Jewelry Stores: 4
Shoe Stores: 3
Department Stores: 1
Book Stores: 3
Clothing Stores: 2

🏬 New Rent Received (7.5% of Profit): $125,250.00
```

Here we observe that increasing the maximum limit has made impact and the rent has gone up significantly. The store counts that changed from the initial constraint were:

**Jewelry (From 2 to 4):**

4 stores at $70k each = $280,000 total whereas 2 stores at $80k = $160,000 previously**.**

Profits here increase relatively well with increased count even though profit per store decreases with more stores according to the excel.

**Department (From 2 to 1) :**

Here the profit decreases because 2 stores gave $420,000 and 1 store gives $270,000.

**Book (From 2 to 3):**

2 stores gave $180,000 while 3 stores give 340,000. So books have a moderate profit

**Conclusion:**

From this we can say that Jewelry and Book stores seem to utilize space more profitably than department stores**.** If space can be increased further**,** adding more Jewelry and Book stores could further maximize profits**.**

**3. The Cubs**

Based on the given restrictions, The Cubs should sign the below pitchers.
This is with respect to the given constraint of spending at most $55Million.

```
Optimal Pitcher Selection and Victory Analysis:
Pitcher: RS, Victories Added: 6, Cost: $25M
Pitcher: BS - Not Selected
Pitcher: DE, Victories Added: 3, Cost: $10M
Pitcher: ST, Victories Added: 3, Cost: $8M
Pitcher: TS, Victories Added: 2, Cost: $7M

 Analysis Summary:
Total Victories Added: 14
Total Cost of Signed Pitchers: $50M
Maximum Achieved Victories (Objective): 14.00
```

We did analyze how would a “change in cost spent” would affect the number of victories and players selected.

```
Budget (M) Selected Pitchers                        Total Victories Total Cost (M)
==========================================================================================
50         RS, DE, ST, TS                           14              50
55         RS, DE, ST, TS                           14              50
60         RS, BS, ST, TS                           16              58
65         RS, BS, ST, TS                           16              58
70         RS, BS, ST, TS                           16              58
```

We see that there is a change in pitchers being selected. And there are 2 victories being added for an budget of $60Million. So is this $10Million dollars for 2 victories worth it?

**Conclusion:**

The management could decide this by considering many factors such as at what stage of the league this decision is being made. If it is at a stage of play offs and these two additional wins could help them win the league, then raising the budget from $50M to $60M would be prudent.

If not, and this is still the very beginning of a league match then the additional $10M investment would not be cost efficient.